

Single-Clock Metrology and the Three-Clock Experiment

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Abstract

The Scientific Reports paper “A single-clock approach to fundamental metrology” (Sci. Rep. 2024, DOI: 10.1038/s41598-024-71907-0) investigates to what extent a single time standard is sufficient as a starting point to define and measure all physical quantities (time intervals, lengths, masses). A central ingredient is an explicit relativistic measurement protocol in which lengths are determined solely from time differences. In addition, the authors argue, using standard quantum relations (Compton wavelength) and modern metrological techniques (Kibble balance), that masses can also be traced back to the time standard.

This document gives a factual summary of the main technical elements of the article and relates them to the T0 theory. In particular, it compares the results to those of the existing T0 documents T0_SI_En, T0_xi_origin_En and T0_xi-and-e_En, where the reduction of all constants to the single parameter ξ and the time–mass duality have already been developed. A short remark on the popular-science video by Hossenfelder places that video as a secondary summary, not as a primary source.

.1 Introduction

The article *A single-clock approach to fundamental metrology* [1] aims at reformulating the foundations of metrology in such a way that a single time standard is sufficient to define all other physical quantities. The authors in particular consider:

- the definition and realization of time intervals by means of a single, highly stable time standard (a “clock”),
- the derivation of length measurements from purely temporal observational data in a relativistic setting,
- the reduction of masses to frequencies or time intervals using established quantum mechanical and metrological relations.

A popular-science presentation of this work appears in a video by Hossenfelder [2]. For the physical argument, however, only the scientific article is decisive; the video is mentioned here for orientation only.

In the T0 theory, T0_SI_En develops a comprehensive derivation scheme in which all fundamental constants and units are obtained from a single geometric parameter ξ . In T0_xi_origin_En and T0_xi-and-e_En, the time–mass duality is analyzed and the internal structure of the mass hierarchy is derived from ξ . The purpose of the present document is to systematically compare these T0 results with the conclusions of the Scientific Reports article.

.2 Time standard and basic assumptions of the article

A single time standard

In the Scientific Reports paper, the starting point is a single, high-precision time standard. Operationally, this means that a reference frequency ν_0 is specified, whose period $T_0 = 1/\nu_0$ defines the elementary unit of time. All other time intervals are given as multiples of T_0 :

$$\Delta t = n T_0, \quad n \in \mathbb{Z}. \quad (1)$$

The concrete physical realization (e.g. caesium atomic clock, optical lattice clock) is left open; what matters is the existence of a stable reference process.

This basic assumption is directly analogous to the T0 theory, where the Planck time t_P and the sub-Planck scale $L_0 = \xi l_P$ are introduced as characteristic scales determined by ξ (T0_SI_En). T0 goes further in that it derives the underlying time structure itself from ξ , while the Scientific Reports article merely assumes the existence of a time standard compatible with known physics.

Relativistic framework

The paper embeds the measurement procedures into special relativity. The key roles are played by:

- proper times of moving clocks along specified worldlines,
- relations between proper time, coordinate time and spatial distance according to the Minkowski metric,
- invariance of the light cone, which constrains the structure of space-time relations.

Formally, the proper time $d\tau$ of an idealized point particle with four-velocity u^μ in flat space-time can be written as

$$d\tau^2 = dt^2 - \frac{1}{c^2} d\vec{x}^2 \quad (2)$$

(with a suitable choice of units). The concrete measurement protocols in the article use this structure to infer spatial separations from measured proper times.

.3 Length measurement from time: three-clock construction

Principle of the procedure

The Nature article analyzes a type of experiment that is conceptually equivalent to the three-clock set-up described by Hossenfelder. The central idea is as follows:

- Two spatially separated events (the ends of a rigid rod) are separated by an unknown distance L .
- Clocks are transported along known worldlines between these points.
- The proper times accumulated by the transported clocks are finally compared at one location.

The authors show that from the proper times of the transported clocks and the known kinematic conditions (e.g. constant speed) one can obtain an equation of the form

$$L = F(\{\Delta\tau_i\}), \quad (3)$$

where $\{\Delta\tau_i\}$ denotes a finite set of measured proper time differences and F is a function determined by special relativity. The crucial point is that F does not require any independently measured length unit.

Operational interpretation

Operationally, this implies that a spatial distance L can in principle be fully determined from times:

$$L = n_L T_0 c_{\text{eff}}. \quad (4)$$

Here T_0 is the elementary time standard, n_L is a dimensionless number obtained from the proper-time measurements and knowledge of the dynamics, and c_{eff} is an effective velocity parameter which, while formally being the speed of light, is not introduced as a separate base quantity. The article emphasizes that no second, independent dimension (a separate meter standard) is needed; the length scale follows from the time structure and the dynamics.

This is consistent with the derivation given in T0_SI_En, where the meter in SI is defined via c and the second, and where c itself is derived from ξ and Planck scales. In T0, therefore, the length unit

is already reduced to the time structure before the metrological construction begins.

.4 Mass determination from frequencies and time

Elementary particles: Compton relation

For elementary particles, the article uses the well-known Compton relation

$$\lambda_C = \frac{\hbar}{mc}, \quad (5)$$

and the corresponding Compton frequency

$$\omega_C = \frac{mc^2}{\hbar}. \quad (6)$$

If lengths have already been defined by time measurements (as in the previous section), it follows that the Compton wavelengths and the masses are also fixed by the time standard. In natural units ($\hbar = c = 1$) this reduces to

$$\lambda_C = \frac{1}{m}, \quad \omega_C = m. \quad (7)$$

Thus mass is a frequency quantity, i.e. an inverse time.

In the T0 theory, this observation appears explicitly in T0_xi-and-e_En in the form

$$T \cdot m = 1. \quad (8)$$

There it is shown that the characteristic time scales of unstable leptons are consistent with their masses once T is taken as a characteristic time and m as mass in natural units. The argument of the Nature article regarding mass determination via frequency measurements therefore finds, within T0, a pre-existing formal elaboration.

Macroscopic masses: Kibble balance

For macroscopic masses, the Nature paper refers to the Kibble balance. This device essentially operates in two modes:

- a static mode, in which the weight force mg of a mass in the gravitational field is balanced by an electromagnetic force,

- a dynamic mode, in which induced voltages and currents are related to quantized electric effects and, finally, to frequencies.

By exploiting quantized electrical effects (Josephson voltage standards, quantum Hall resistances), one obtains a chain

$$m \longrightarrow F_{\text{weight}} \longrightarrow U, I \longrightarrow \text{frequencies, counting} \longrightarrow T_0. \quad (9)$$

Formally, the mass m is thereby reduced to a function of frequencies (time standards) and discrete charge counts. Again, no new continuous base quantities appear; electrical and thermal constants are coupled to the time norm via defining relations.

In T0, T0_SI_En derives the corresponding relations for e , α , k_B and further constants from ξ , so that the Kibble balance can be interpreted as an experimental realization of an already geometrically fixed constants network.

.5 Relation to the T0 documents

T0_SI_En: From ξ to SI constants

T0_SI_En presents in detail how, starting from the single parameter ξ , one can derive the gravitational constant G , Planck length l_P , Planck time t_P and finally the SI value of the speed of light c . The central relation

$$\xi = 2\sqrt{Gm_{\text{char}}} \quad (10)$$

and its variants ensure consistency with CODATA values and with the SI 2019 reform.

Against this background, the single-clock metrology of the Scientific Reports paper can be interpreted as follows:

- The claim that a single time standard suffices is consistent with the T0 statement that ξ as a single fundamental parameter suffices.
- The reduction of SI units to time and counting units mirrors the T0 description of reducing all constants to ξ .

T0_xi_origin_En: Mass scaling and ξ

T0_xi_origin_En addresses how the concrete numerical value $\xi = 4/30000$ emerges from the structure of the e-p- μ system, the

fractal space-time dimension and related considerations. This internal justification level is absent from the Scientific Reports article: there, one simply assumes that a time standard exists and can be reconciled with known physics.

From the T0 perspective, the mass–frequency relation used in the article is therefore not only accepted, but traced back to a deeper geometric level in which mass ratios appear as consequences of ξ . The metrological statement of the paper is thereby supported and at the same time embedded into a broader theoretical framework.

T0_xi-and-e_En: Time–mass duality

In T0_xi-and-e_En, the relation $T \cdot m = 1$ is highlighted as an expression of a fundamental time–mass duality. The Scientific Reports article uses this duality in the form of established relations (Compton wavelength, mass–frequency relation) without explicitly formulating it as a duality.

The comparison shows:

- The article uses the duality operationally to argue that masses can be fixed by a time standard.
- The T0 theory formulates the duality explicitly and anchors it in the geometric structure (parameter ξ) and in the mass hierarchy of the particles.

.6 Quantum gravity and range of validity

The Nature article formulates its claims within the framework of established physics, i.e. based on special relativity, quantum mechanics and the current metrological standard model. Hossenfelder points out that the argument implicitly assumes that clocks can, in principle, be used with arbitrarily high precision. In the regime of Planck scales this expectation will likely fail, since quantum-gravitational effects should lead to fundamental uncertainties.

The T0 theory addresses this issue by introducing Planck length, Planck time and the sub-Planck scale as quantities determined by ξ . In T0_SI_En, $L_0 = \xi l_P$ is discussed as an absolute lower bound of space-time granulation. Planck scales thereby appear in T0 not as additional parameters independent of ξ , but as derived quantities.

In this sense, the domain of validity of the single-clock metrology argument can be characterized as follows:

- Within the T0-described range (above L_0 and t_P), the reduction to a single time standard is consistent with the geometric structure.
- Below these scales, a modification of the measurement concept is to be expected; single-clock metrology does not provide a complete answer in this regime, and T0 proposes a concrete structure of these sub-Planck scales.

.7 Concluding remarks

The Scientific Reports article on single-clock metrology shows that a consistent use of special relativity, quantum mechanics and modern metrology leads to the result that a single time standard is, in principle, sufficient to define and measure all physical quantities. Length measurement from time differences (three-clock construction) and mass determination via frequencies and Kibble balances are the central technical building blocks.

The T0 theory, especially in T0_SI_En, T0_xi_origin_En and T0_xi-and-e_En, provides a complementary viewpoint in which these operational facts are traced back to a single geometric parameter ξ . Time is the primary quantity; mass appears as inverse time, and all SI constants are derived from ξ or interpreted as conventions. The single-clock metrology of the article can thus be viewed as a metrological confirmation of the time–mass duality and single-parameter structure postulated in T0.

Bibliography

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